

Dissecting the illocutionary force phrase: New evidence from SFP embeddability

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Abstract: This paper focuses on the peripheral hierarchy of Mandarin interrogatives. Based on question markers' embeddability profiles in a series of non-quotational subordinating contexts, this paper argues for a three-way split in projections that contribute to the illocutionary force, attributing the non-co-occurrence of question markers to their semantics. Specifically, we propose that (i) LowIForceP houses the most embeddable *meiyou*, Op_{WH} , and A-not-A; (ii) PerspP houses question markers restrictedly embeddable like *bu* and *ma*; and (iii) HighIForceP houses the unembeddable *ba*. Looking deeper into the embeddability patterns of *bu* and *ma*, we argue that our PerspP should be interpreted à la Dayal 2023 in the sense that it introduces a not-at-issue requirement for the embedded question to be potentially active for the matrix subject. We explicate that the embedding of *bu* and *ma* is restricted by two constraints: (i) the embedded clause size must satisfy the syntactic selectional requirement of the embedding context; and (ii) the overall interpretation must be compatible with a not-at-issue requirement lexically introduced. Our proposal calls for a theory of selection that manifests a conspiracy of syntax and semantics.

Keywords: *question marker, sentence-final particle, left periphery, illocutionary force phrase, perspective phrase, Mandarin Chinese*

1 Introduction

Sentence-final particles (SFPs) occupy a significant place in Chinese linguistic research, with *ba* and *ma* notable for marking questions. In interrogatives, *ba* denotes confirmation questions, typically translated into English as tag questions, and *ma* denotes (unbiased) yes-no

questions (a.o.: Li & Thompson 1981, Cheng 1991). Another sentence-final question marker — though typically not considered an SFP — *meiyou*, also denotes (unbiased) yes-no questions but conveys the perfective aspect with or without the perfective suffix (a.o.: Pan 2015). Examples are given in (1–3) along with the declarative base in (4).

- | | |
|--|--|
| <p>(1) Zuotian xia-le yu ba?
yesterday fall-PFV rain BA
'It rained yesterday, didn't it?'</p> | <p>(2) Zuotian xia-le yu ma?
yesterday fall-PFV rain MA
'Did it rain yesterday?'</p> |
| <p>(3) Zuotian xia(-le) yu meiyou?
yesterday fall-PFV rain MEIYOU
'Did it rain yesterday?'</p> | <p>(4) Zuotian xia-le yu.
yesterday fall-PFV rain
'It rained yesterday.'</p> |

Meiyou, *ma*, and *ba* are traditionally placed within the same syntactic projection, such as Paul's (2014b) C₂P or Pan's (2015) *illocutionary force phrase* (iForceP), a perspective motivated by their complementary distribution. In this paper, we advocate for a separate layer for each of these question markers, and we attribute their non-co-occurrence to semantic mismatches as opposed to syntactic competitions. Our proposal understands selectional restrictions as a conspiracy of both syntax and semantics and extends to *all* Mandarin question markers (to the extent that they may be called question markers), using the question markers' embeddability patterns as a window.

Section 2 reviews the traditional view on the Mandarin peripheral hierarchy and presents newly surfaced data that challenge this view, calling for a new analysis. Sections 3 to 4 lay out the two components of our analysis, theorizing selection as a conspiracy of syntax and semantics. Section 5 concludes.

2 Traditional view: One head

Since *meiyou*, *ma*, and *ba* never co-occur with each other, they are proposed to realize the same functional head:

- (5) a. Paul 2014b (inspired by Zhu 1982)
low complementizers < force complementizers < attitude complementizers
- b. Pan 2015 *et seq.*
SAspP < OnlyP < iForceP < SQP < AttitudeP

Against this background, in this section, we review prior observations and analyses of these question markers' (un)embeddability. We then present newly surfaced data that challenge these observations and call for a revised analysis.

2.1 Earlier work on the question markers' (un)embeddability

Earlier work (a.o.: Zhu 1982, Paul 2014a,b, Pan 2015, 2019) observes that *meiyou* can always embed, while *ma* and *ba* never embed. This observation is illustrated using noun complements and clausal subjects resembling (6).

- (6) [Zuotian xia yu le meiyou/*ma/*ba] (de wenti) hen guanjian.
 Yesterday fall rain PRF MEIYOU MA BA COMP question very crucial
 (Intended:) ‘(The question of) whether it rained yesterday is very crucial.’

This observation is accounted for by the lexical properties of question markers. For instance, Paul (2014b) proposes that SFPs carry a $[\pm\text{root}]$ feature, and only $[-\text{root}]$ SFPs are embeddable. Under this proposal, *ma* and *ba* are $[\text{+root}]$, and while Paul does not discuss *meiyou*, given its embeddability, it would be considered $[-\text{root}]$.

It is therefore predicted that the unembeddability of *ba* and *ma* should (i) share exactly the same pattern because they are both $[\text{+root}]$ and (ii) remain intact across different embedding context – in particular, they should be insensitive to the properties of matrix predicates when occurring in clausal objects. However, as we show in the next subsection, when newly surfaced data are taken into consideration, these two predictions are not borne out.

2.2 Challenges from new data

Predictions made by the lexical account are challenged by Bhatt & Dayal’s (2020) data: *ma* is embeddable after all. We copy their example in (7) and use “↑” to represent the intonation typically associated with Mandarin information-seeking yes-no questions.

- (7) (Bhatt & Dayal 2020: 1122)
 John xiang zhidao [xia yu le **ma** zuotian]↑.
 John want know fall rain PRF MA yesterday
 ‘John wants to know whether it rained yesterday.’

Prompted by Bhatt & Dayal, we add in (8) examples involving pronoun coreference and variable binding, which are evidence that the questions are not quotations and are truly embedded. If they were quotations, personal pronouns would behave differently (McCloskey 2006), as in (9).

- (8) *Embedded questions: 1SG = speaker and 3SG = matrix subject*
- a. Xiaoming_i xiang zhidao [ta_i de fenshu bi wo_j gao **ma**]↑.
 Xiaoming want know 3SG POSS score than 1SG high MA
 ‘Xiaoming_i wants to know whether his_i score is higher than mine_j.’
 - b. [Mei ge ren]_i dou xiang zhidao [ta_i de fenshu bi wo_j gao **ma**]↑.
 every CLF person all want know 3SG POSS score than 1SG high MA
 ‘Everyone_i wants to know whether their_i score is higher than mine_j.’
- (9) *Quoted questions: 1SG = quoting predicate subject and 3SG = someone else*
- a. Xiaoming_i xiang zhidao: [“ta_j de fenshu bi wo_i gao **ma**?”]
 Xiaoming want know 3SG POSS score than 1SG high MA
 ‘Xiaoming_i wants to know, “is his_j score higher than mine_i?”’
 - b. [Mei ge ren]_i dou xiang zhidao: [“ta_j de fenshu bi wo_i gao **ma**?”]
 every CLF person all want know 3SG POSS score than 1SG high MA
 ‘Everyone_i wants to know, “is their_j score higher than mine_i?”’

Utterances like that in (8a) can be felicitously made in a context in which the speaker acts as a proxy for the matrix subject, posing a question on their behalf and declaring their curiosity about the answer. An example is (10).

(10) *Context for (8a)*

(Xiaoming and John both took a seminar by Mary. The grades had not been posted yet, and Xiaoming wanted to know whether he or John got a higher grade. Being a very shy student, he did not want to ask Mary himself. He asked John to ask Mary. So, John said to Mary, “Xiaoming wants to know whether his score is higher than mine.”)

Data resembling (8) have been confirmed in Beijing Mandarin under experimental conditions by Liu (to appear) using a novel semantic judgment elicitation paradigm. While we believe that the acceptability is found in a much wider range of dialects, we acknowledge regional variations, which we leave for future work.

The embeddability of *ma* depends heavily on the matrix predicate type. While predicates like ‘want to know’ accept it, other predicates like ‘know’ reject it:

- (11) #Xiaoming zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaoming know 3SG POSS score than 1SG high MA
 Intended: ‘Xiaoming_i knows whether his_i score is higher than mine_j.’

We find that this contrast between ‘want to know’ and ‘know’ does not align with the familiar rogative-responsive distinction (building on Dayal’s (2023) observations of English embedded inverted questions). This can be seen from predicates like ‘forget’, which is responsive but patterns with ‘want to know’ instead of ‘know’, as it allows the embedding of *ma* (12).

- (12) Xiaoming_i wangji-le [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaoming forget-PFV 3SG POSS score than 1SG high MA
 ‘Xiaoming_i forgot whether his_i score is higher than mine_j.’

The nature of this contrast has to do with whether the predicate is semantically compatible with a context in which the possibility that the matrix subject does not know the answer to the embedded question exists: *ma* necessitates the manifestation of this possibility in the context, a condition we refer to as *ignorance requirement* (IR). We follow Dayal & Grimshaw (2009) and term a predicate that satisfies the IR (e.g., ‘want to know’) a *quasi subordinator* (QS) and one that does not (e.g., ‘know’) a non-quasi subordinator (NQS). An illustration of the contrast between QS and NQS in terms of the IR as well as a comparison with rogative and responsive predicates is presented in Table 1.

	QS (✓IR): ◇¬know(<i>x</i> , answer(<i>Q</i>))	NQS (✗IR): □know(<i>x</i> , answer(<i>Q</i>))
Rogative: [+Q]	<i>care about, want to know, wonder</i>	
Responsive: [±Q]	<i>forget</i>	<i>know, remember</i>

Table 1 Question embedding predicates of different types

Unlike *ma*, *meiyou* and *ba*'s (un)embeddability is unaffected by the QS-NQS contrast and aligns with the traditional view: *meiyou* is always embeddable, while *ba* never is (although *ba* can occur in quotations). We thus observe a trichotomy in the (un)embeddability of question markers, which challenges binary features like $[\pm\text{root}]$ and prompts a new analysis.

3 First component: Split iForceP

This section extends the observed trichotomy to all Mandarin question markers and argues that Dayal's (2023) three-layer peripheral structure is systematically manifested in Mandarin.

3.1 Extending the trichotomy to all question markers

Other Mandarin question markers in iForceP that Pan (2015) et seq. identifies include the SFP *bu* and the *wh*-question operator. Additionally, A-not-A questions are proposed to be marked by a covert clause-typing head (e.g., Huang 1982), whose embeddability merits examination.

Bu signals yes-no questions when following a declarative base (Pan 2019). We observe that *bu* questions exhibit the same embeddability pattern as *ma* questions: it is embeddable in a clausal object of a QS but unembeddable elsewhere (13).

- (13) a. *Noun complement or clausal subject*
 * [Nuobei'er Jiang hui ban gei ta bu] (de wenti) hen guanjian.
 Nobel Prize will award to 3SG BU COMP question very crucial
 Intended: '(The question of) whether the Nobel Prize will be awarded to him is very crucial.'
- b. *Clausal object of an NQS*
 # Zhangsan_i zhidao [Nuobei'er Jiang hui ban gei ta_i bu]_↑.
 Zhangsan know Nobel Prize will award to 3SG BU
 Intended: 'Zhangsan_i knows whether the Nobel Prize will be awarded to him_i.'
- c. *Clausal object of a QS*
 Zhangsan_i xiang zhidao [Nuobei'er Jiang hui ban gei ta_i bu]_↑.
 Zhangsan want know Nobel Prize will award to 3SG BU
 'Zhangsan_i wants to know whether the Nobel Prize will be awarded to him_i.'

In contrast, *wh*-questions are embeddable in all embedding contexts and pattern like *meiyou*. As for A-not-A, it signals yes-no questions and is formed by reduplicating either the verb or its leftmost portion and placing a negative morpheme in between. As presented in (14), A-not-A is also embeddable across the board, resembling *meiyou*.

- (14) a. *Noun complement or clausal subject*
 [Mingtian xia-bu-xia yu] (de wenti) hen guanjian.
 tomorrow fall-not-fall rain COMP question very crucial
 '(The question of) whether it will rain tomorrow is very crucial.'
- b. *Clausal object of an NQS*
 Zhangsan zhidao [mingtian xia-bu-xia yu].
 Zhangsan know tomorrow fall-not-fall rain
 'Zhangsan knows whether it will rain tomorrow.'

- c. *Clausal object of a QS*
 Zhangsan xiang zhidao [mingtian xia-bu-xia yu].
 Zhangsan want know tomorrow fall-not-fall rain
 ‘Zhangsan wants to know whether it will rain tomorrow.’

The embeddability patterns covered thus far are summarized in Table 2.

	<i>Meiyou</i>	Op_{WH}	A-not-A	<i>Bu</i>	<i>Ma</i>	<i>Ba</i>
Noun complement	✓	✓	✓	✗	✗	✗
Clausal subject	✓	✓	✓	✗	✗	✗
Clausal object of an NQS	✓	✓	✓	✗	✗	✗
Clausal object of a QS	✓	✓	✓	✓	✓	✗

Table 2 Embeddability of question markers

3.2 Split iForceP proposal

We capture the observed tiered embeddability pattern using a three-way split iForceP (cf. Dayal’s (2023) CP, PerspP, and SAP) and label the projections as low illocutionary force phrase (LowIForceP), perspective phrase (PerspP), and high illocutionary force phrase (HighIForceP). LowIForceP accommodates the most embeddable elements, including *meiyou*, the *wh*-question operator, and A-not-A. PerspP accommodates elements embeddable under restricted conditions, including *bu* and *ma* (cf. Luo to appear). HighIForceP accommodates the unembeddable element *ba*. The revised periphery looks like (15).

- (15) *Mandarin peripheral hierarchy under the split iForceP proposal*
 SASpP < OnlyP < LowIForceP < PerspP < HighIForceP < SQP < AttitudeP
- LowIForceP: *meiyou*, Op_{WH} , and A-not-A
 - PerspP: *bu* and *ma*
 - HighIForceP: *ba*

3.3 Semantic account for non-co-occurrence

The split iForceP proposal phases out the syntactic competition between the aforementioned question markers and calls for a new analysis of their complementary distribution. We attribute their non-co-occurrence to their semantics: the output of a lower question marker is not of the right type to serve as an input of any higher question markers.

To concretely spell out our proposal, we adopt a Karttunen (1977) analysis for question semantics and assume that a question is of type $\langle st, t \rangle$ (i.e., a set of propositions) and has a propositional nucleus of type *st*. Specifically, we propose that *meiyou*, A-not-A, *bu*, and *ma* take propositions and yield questions, while we follow Ye’s (2024) proposal that *ba* takes and yields propositions. Thus, since all question markers take propositions, while all question

markers (except for *ba*, which belongs to the highest projection and cannot feed its output to a lower question marker) yield sets of propositions, they cannot co-occur with each other due to type clash. Our full proposal for their semantics is summarized in Table 3. Deferring a detailed discussion of the semantic types to the next subsection, we note that the non-co-occurrence can be explained under our proposal as long as we assume that propositions and questions have different semantic types.

LowIForceP	PerspP	HighIForceP
$p \rightarrow \text{meiyou} \rightarrow \{p, \neg p\}$	$p \rightarrow \text{bu} \rightarrow \{p, \neg p\}$	$p \rightarrow \text{ba} \rightarrow p$
$p \rightarrow \text{A-not-A} \rightarrow \{p, \neg p\}$	$p \rightarrow \text{ma} \rightarrow \{p\}$	

Table 3 Semantic types of the input and output of question markers

3.4 Motivation for assigning different output semantic types

As shown in Table 3, we go even further and specify that *ma* yields a singleton, while all other question markers except for *ba* yield doubletons. This singleton-doubleton distinction naturally accounts for three distributional facts: the SFP *a*, the bias marker *nandao*, and ‘yes’ as a response.

The SFP *a* (which may be realized as *na*, *nga*, *ra*, *wa*, *ya*, and *za* in different phonological contexts) conveys the speaker’s attitude and emotion (Zhu 1982). When used in a question, we informally describe its semantic or pragmatic role as strengthening the interrogative speech act. It cannot occur with *ma* but can occur with all other question markers except for *ba* (16). It can also occur with *wh*-questions. This distribution can be naturally captured by positing that *a* semantically selects nonsingletons.

- (16) a. (Adapted from Zhu 1982: 203)
 Baozhi lai-le **meiyou/bu/*ma/*ba** a?
 newspaper come-PFV MEIYOU BU MA BA A
 (Intended:) ‘Did the newspaper arrive?’
- b. Baozhi **lai-bu-lai** /**lai-mei-lai** a?
 newspaper come-not-come come-not.PFV-come A
 ‘Will/did the newspaper arrive?’

Nandao triggers a negative bias toward the question nucleus (Xu 2017). It can occur with *ma* and no other question markers (17). It cannot occur with *wh*-questions, either. This distribution can be naturally captured by positing that *nandao* semantically selects singletons.

- (17) a. (Adapted from Xu 2017: 13)
Nandao Zhangsan chi-le fan **ma/*meiyou/*bu/*ba**?
 NANDAO Zhangsan eat-PFV meal MA MEIYOU BU BA
 (Intended:) ‘Zhangsan didn’t eat, right?’

- b. *Nandao Zhangsan **chi-bu-chi**/**chi-mei-chi** fan?
 NANDAO Zhangsan eat-not-eat eat-not.PFV-eat meal
 Intended: ‘Zhangsan don’t/didn’t eat, right?’

Finally, only *ma* questions and confirmation questions with *ba* can be felicitously answered with *dui*, *mei cuo*, or *shi de*, all of which roughly correspond to English *yes*. This distribution can be naturally captured by positing that ‘yes’ only responds to singletons and propositions (cf. *propositional anaphor* in the sense of Krifka 2013). The distributions of *nandao*, *a*, and ‘yes’ as a response are summarized in Table 4 and motivate the output semantic types we designate in Table 3.

	<i>Meiyou</i> $\{p, \neg p\}$	Op_{WH} $\{\phi(x_1), \dots, \phi(x_n)\}$	A-not-A $\{p, \neg p\}$	<i>Bu</i> $\{p, \neg p\}$	<i>Ma</i> $\{p\}$	<i>Ba</i> p
<i>A</i>	✓	✓	✓	✓	✗	✗
<i>Nandao</i>	✗	✗	✗	✗	✓	✗
‘Yes’ as a response	✗	✗	✗	✗	✓	✓

Table 4 Distributions of *a*, *nandao*, and ‘yes’ as a response

4 Second component: Conspiratorial selection

The second component of our analysis argues that PerspP should be interpreted à la Dayal 2023 in the sense that it triggers perspectival centering and introduces a requirement on the embedded question’s potential activeness for the perspectival center. We show how *bu* and *ma* align with Dayal’s prediction that their embeddability is sensitive to the matrix negation and illocutionary force in addition to the matrix predicate type.

4.1 P-activeness

Dayal (2023) proposes that Persp^0 poses a not-at-issue requirement that the complement question must be potentially active (p-active) for the perspectival center. Two concepts merit further clarification: *perspectival center* and *p-activeness*.

A perspectival center, roughly speaking, is “the one who holds the question.” In Dayal’s implementation, the perspectival center is syntactically introduced in SpecPerspP by a phonologically null anaphor (i.e., PRO) that must be bound. In a matrix question, this anaphor is bound by the speaker. When the PerspP is embedded (18), the anaphor is bound by an argument of the matrix predicate, typically the matrix subject.

- (18) *Perspectival center = matrix argument* (adapted from Bhatt & Dayal 2020: 1122)
 (The question of whether it rained is p-active for the matrix subject John.)

John_i xiang zhidao [_{PerspP} PRO_i [_{Persp'} xia yu le [_{Persp⁰} ma]]]↑.
 John want know fall rain PRF MA
 ‘John wants to know whether it rained.’

For a question Q to be p-active for the perspectival center x , a minimal requirement is the possibility of x 's ignorance about the answer to Q (19), or the IR as discussed in Section 2.2. This not-at-issue requirement is formalized as a lexical property of Persp^0 , as in (20).

- (19) *Minimal requirement for p-activeness* (Dayal 2023: 11)
 $\forall Q [\forall x [\text{p-active-for}(Q, x) \rightarrow \Diamond \neg \text{know}(x, \text{answer}(Q))]]$
- (20) *Denotation of Persp^0* (Dayal 2023: 11)
 $[\text{Persp}^0] = [\lambda Q. \lambda x : \text{p-active-for}(Q, x). Q]$

Note that in Dayal's framework, Persp^0 takes a question as its input. That is, clause-typing takes place uniformly in a separate projection below PerspP . For Persp^0 in Mandarin (i.e., *bu* and *ma*), we slightly modify their denotation by allowing them to take a proposition as input. *Bu* and *ma* return a doubleton (21a) and a singleton (21b), respectively. We use $\text{PR}(x)$ to abbreviate the p-activeness requirement for x .

- (21) a. $[\text{bu}] = [\lambda p. \lambda x : \text{PR}(x). [\lambda q. q = p \vee q = \neg p]]$
 b. $[\text{ma}] = [\lambda p. \lambda x : \text{PR}(x). [\lambda q. q = p]]$

The paradigmatic distinctions between QS and NQS in their embedding of *bu* and *ma* follow from the PR. Since *bu* and *ma* realize Persp^0 , they introduce the PR when embedded, with the matrix subject being the perspectival center. The PR cannot be satisfied by an NQS due to its semantics, whereas it can be met by a QS:

- (22) *NQS: ✗PR*
 #John zhidao [xia yu le ma]_↑.
 John know fall rain PRF MA
 Intended: 'John knows whether it rained.'
 a. John knows the answer to the question of whether it rained.
 b. It is not possible that John does not know the answer.
 c. The question of whether it rained is not p-active for John.
- (23) *QS: ✓PR* (adapted from Bhatt & Dayal 2020: 1122)
 John xiang zhidao [xia yu le ma]_↑.
 John want know fall rain PRF MA
 'John wants to know whether it rained.'
 a. John wants to know the answer to the question of whether it rained.
 b. It is possible that John does not know the answer.
 c. The question of whether it rained is p-active for John.

Concretely, we propose that question embedding predicates, be they QS or NQS, can freely select a LowIForceP or a PerspP. There is nothing in the syntax that prevents a PerspP from being the complement of a predicate of either type. Yet, an NQS is incompatible with contexts in which the centered question is p-active for the perspectival center: one cannot possibly be ignorant about the answer to a question when they already know it.

4.2 Evidence for semantic selection

The presence of the PR demands a conspiratorial theory of selection, where syntax and semantics both contribute to the (un)acceptability of clausal complementation. A piece of evidence for semantic selection comes from the observation originally noted by McCloskey (2006) that the availability of quasi subordination is sensitive to the matrix negation and illocutionary force in addition to the matrix predicate type, conforming with predictions by the PR. When the matrix clause itself changes from an affirmative declarative to a negative declarative or an interrogative, the (in)compatibility of the matrix predicate with a p-active context may shift. To illustrate, take again the NQS ‘know’ for example. As discussed above, a statement of “ x knows Q ” blocks the possibility of x ’s ignorance about the answer to Q , but a statement of “ x does not know Q ” or a question of “Does x know Q ?” keeps this possibility open, since it becomes possible that x does not know Q . Consequently, when the matrix clause is negative or interrogative, quasi subordination is predicted to be acceptable even when embedded under an NQS. Liu (to appear) provides quantitative data to show that this prediction is confirmed in Mandarin. Some examples are given below:

- (24) a. *Affirmative declarative: #*
 #Xiaoming_i zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaoming know 3SG POSS score than 1SG high MA
 ‘Xiaoming_i knows whether his_i score is higher than mine_j.’
 b. *Negative declarative: OK*
 Xiaoming_i bu zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaoming not know 3SG POSS score than 1SG high MA
 ‘Xiaoming_i does not know whether his_i score is higher than mine_j.’
 c. *Interrogative: OK*
 Xiaoming_i zhi-bu-zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaoming know-not-know 3SG POSS score than 1SG high MA
 ‘Does Xiaoming_i know whether his_i score is higher than mine_j?’

Additionally, with a possibility adverb, *bu* and *ma* become embeddable under an NQS (Dan-feng Wu, pers. comm.):

- (25) Huoxu/yexu Xiaoming_i zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 maybe maybe Xiaoming know 3SG POSS score than 1SG high MA
 ‘Maybe Xiaoming_i knows whether his_i score is higher than mine_j.’

All four sentences share the same matrix predicate ‘know’, which would presumably impose the same syntactic constraint on its complement, suggesting that the acceptability filter at work comes from semantics (i.e., the PR lexically encoded in *bu* or *ma*) rather than syntax.

Another piece of evidence has to do with the fact that the way to say ‘wonder’ in Mandarin is lexically composed of ‘want’ and ‘know’. Should selection be purely syntactic here, ‘want to know’ would share the same embedding profile as ‘know’. Yet, this is not true, as we have seen in Section 2.2, where QS and NQS differ in their sensitivity to the PR.

4.3 Evidence for syntactic selection

Two questions remain unaddressed: (i) why *bu* and *ma* cannot be embedded in clausal subjects, noun complements, or clausal objects of prepositions; and (ii) why this data pattern necessitates a distinction between LowIForceP and PerspP. The answers to these questions are closely related and present evidence for the need of syntactic selection on par with semantic selection.

Empirically, noun complements & clausal subjects (discussed in Section 2.1) and clausal objects of prepositions cannot host *bu* and *ma*. In the current account, we attribute this observation to their requirements on the complement clause size: a PerspP is too large for them to embed. Therefore, *bu* and *ma* are banned from these contexts, regardless of the epistemic status of the question holder. On the other hand, clausal objects of question embedding predicates in our proposal allow for larger structures (i.e., PerspP), where the PR comes into play, leading to the QS-NQS contrast in embedding *bu* and *ma*.

Crucially, an alternative account that tries to avoid the splitting of PerspP from LowIForceP makes false predictions. Such an account would predict that *bu* and *ma* questions can be embedded as clausal subjects and clausal objects of prepositions, as long as the context satisfies the PR. This prediction is not borne out. The question holders in (26) – namely, the speaker in (26a) and the matrix subject in (26b) – are ignorant of the answer to the question, but the sentences are still unacceptable.

- (26) a. *Noun complement or clausal subject*: ✗
 *[Zuotian xia yu le **ma**] (de wenti) hai bu qingchu.
 yesterday fall rain PRF MA COMP question yet not clear
 Intended: ‘(The question of) whether it rained yesterday is not clear yet.’
- b. *Clausal object of a preposition*: ✗
 *Xiaoming dui [zuotian xia yu le **ma**] hen haoqi.
 Xiaoming about yesterday fall rain PRF MA very curious
 Intended: ‘Xiaoming is very curious about whether it rained yesterday.’

4.4 Derivations of the question markers’ (un)embeddability

With the two components established, we derive the (un)embeddability patterns of all aforementioned question markers. *Meiyou*, Op_{WH}, and A-not-A are all completely embeddable; they do not introduce the PR, and we consider them LowIForce heads. The derivation of their embeddability is illustrated using *meiyou* in (27–28).

- (27) *Embedded meiyong question as noun complement or clausal subject*: ✓
 [Zuotian xia yu le **meiyou**] (de wenti) hen guanjian.
 Zhangsan fall rain PRF COMP question very crucial
 ‘(The question of) whether it rained yesterday is very crucial.’
- a. Noun complements and clausal subjects must be no larger than LowIForceP.
 b. A *meiyou* question is a LowIForceP and not larger than LowIForceP (✓syntax).
 c. (27) is acceptable.

(28) *Embedded meiyou question as clausal object of a question embedding predicate:* ✓Xiaoming_i (xiang) zhidao [ta_i jige meiyou].

Xiaoming want know 3SG pass MEIYOU

‘Xiaoming_i knows/wants to know whether he_i passed.’

- a. Clausal objects of question embedding predicates must be no larger than PerspP.
- b. A *meiyou* question is a LowIForceP and not larger than PerspP (✓syntax).
- c. (28) is acceptable.

As for *bu* and *ma* questions, they are not embeddable as noun complements or clausal subjects due to their size being PerspPs and exceeding LowIForcePs. When embedded as clausal objects, they introduce the PR. We illustrate their derivation using *ma* in (29–31).

(29) *Embedded ma question as noun complement or clausal subject:* ✗

[Zuotian xia yu le ma] (de wenti) hen guanjian.

yesterday fall rain PRF MA COMP question very crucial

Intended: ‘(The question of) whether it rained yesterday is very crucial.’

- a. Noun complements and clausal subjects must be no larger than LowIForceP.
- b. A *ma* question is a PerspP and larger than LowIForceP (✗syntax).
- c. (29) is unacceptable.

(30) *Embedded ma question as clausal object of an NQS (in an affirmative declarative):* ✗Xiaoming zhidao [ta_i de fenshu bi wo_j gao ma]_↑.

Xiaoming know 3SG POSS score than 1SG high MA

Intended: ‘Xiaoming_i knows whether his_i score is higher than mine_j.’

- a. Clausal objects of question embedding predicates must be no larger than PerspP.
- b. A *ma* question is a PerspP and not larger than PerspP (✓syntax).
- c. A *ma* question introduces the PR.
- d. Since the matrix subject knows the answer, the PR is not satisfied (✗semantics).
- e. (30) is unacceptable.

(31) *Embedded ma question as clausal object of a QS (in an affirmative declarative):* ✓Xiaoming xiang zhidao [ta_i de fenshu bi wo_j gao ma]_↑.

Xiaoming want know 3SG POSS score than 1SG high MA

‘Xiaoming_i wants to know whether his_i score is higher than mine_j.’

- a. Clausal objects of question embedding predicates must be no larger than PerspP.
- b. A *ma* question is a PerspP and not larger than PerspP (✓syntax).
- c. A *ma* question introduces the PR.
- d. Since the matrix subject wants to know the answer, the PR is satisfied (✓semantics).
- e. (31) is acceptable.

Ba questions are unembeddable across the board (aside from their use in quotations). We attribute their unembeddability to their exceedingly large clause size and illustrate this in (32–33).

(32) *Embedded ba question as noun complement or clausal subject: ✗*

*[Zuotian xia yu le ba] (de wenti) hen guanjian.
yesterday fall rain PRF BA COMP question very crucial

Intended: ‘(The question of) whether it rained yesterday is very crucial.’

- a. Noun complements and clausal subjects must be no larger than LowIForceP.
- b. A *ba* question is a HighIForceP and larger than LowIForceP (**✗**syntax).
- c. (32) is unacceptable.

(33) *Embedded ba question as clausal object of a question embedding predicate: ✗*

*Xiaoming_i (xiang) zhidao [ta_i de fenshu bi wo_j gao ba]_↑.

Xiaoming want know 3SG POSS score than 1SG high BA

Intended: ‘Xiaoming_i knows/wants to know whether his_i score is higher than mine_j.’

- a. Clausal objects of question embedding predicates must be no larger than PerspP.
- b. A *ba* question is a HighIForceP and larger than PerspP (**✗**syntax).
- c. (33) is unacceptable.

5 Conclusion

We observe a trichotomy in the embeddability profiles of Mandarin question markers: (i) some are always embeddable; and (ii) some are restrictedly embeddable and are sensitive to the QS-NQS contrast; while (iii) some are unembeddable across the board. This trichotomy is best captured if we adopt a theory of selection that manifests a conspiracy of syntax and semantics. Syntactic selection specifies the maximum clause size an embedding context allows, while semantic selection introduces a PR that checks whether the embedded question is p-active for the perspectival center. In this regard, not only does this paper add novel data to the discussion of how questions are built up in the left periphery, but it also contributes to our understanding of the division of labor between different modules of the grammar with respect to selection.

Abbreviations

1 = first person, 3 = third person, CLF = classifier, COMP = complementizer, PFV = perfective, PRF = perfect, POSS = possessive, SG = singular.

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